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(54) **METHOD OF PRODUCING  
 OPTOELECTRONIC SEMICONDUCTOR  
 COMPONENTS AND AN OPTOELECTRONIC  
 SEMICONDUCTOR COMPONENT**

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*H10H 29/142* (2025.01); *H10H 20/0361*  
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(57) **ABSTRACT**

A method of producing an optoelectronic semiconductor component includes providing a primary light source having a carrier and a semiconductor layer sequence mounted thereon that generates primary light, wherein the semiconductor layer sequence is structured into a plurality of pixels driven independently of each other, the carrier including multiple control units that drive the pixels, providing a conversion unit for converting the primary light into secondary light, wherein the conversion unit is grown continuously from semiconductor material, structuring the conversion unit, removing portions of the semiconductor material in accordance with the pixels, and applying the conversion unit to the semiconductor layer sequence so that the remaining semiconductor material is uniquely assigned to a portion of the pixels, wherein at least two conversion units are grown on a common growth substrate, and wherein at least one of the conversion units is structured while still applied to the common growth substrate.

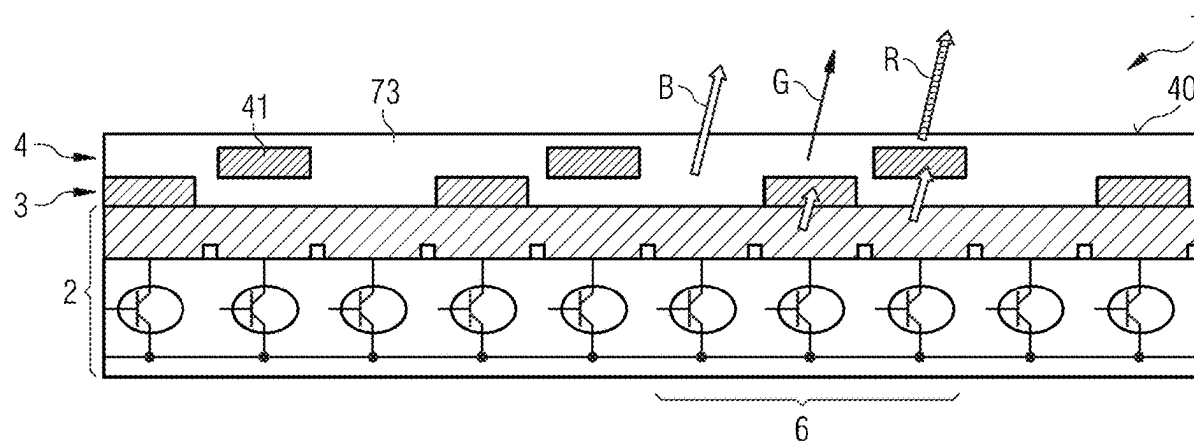


FIG 1A

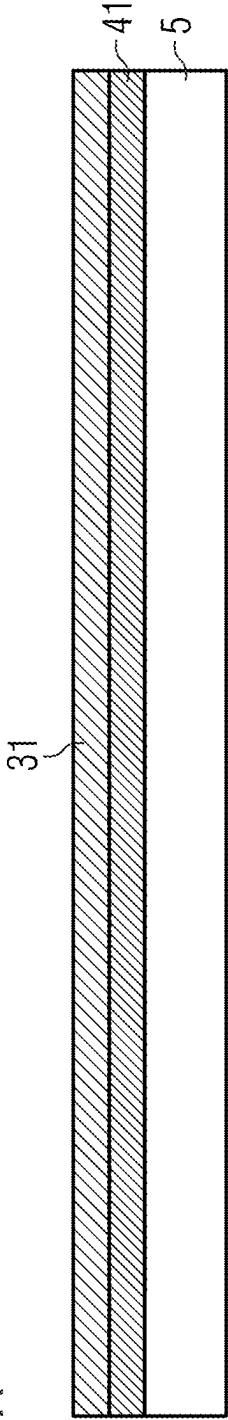


FIG 1B

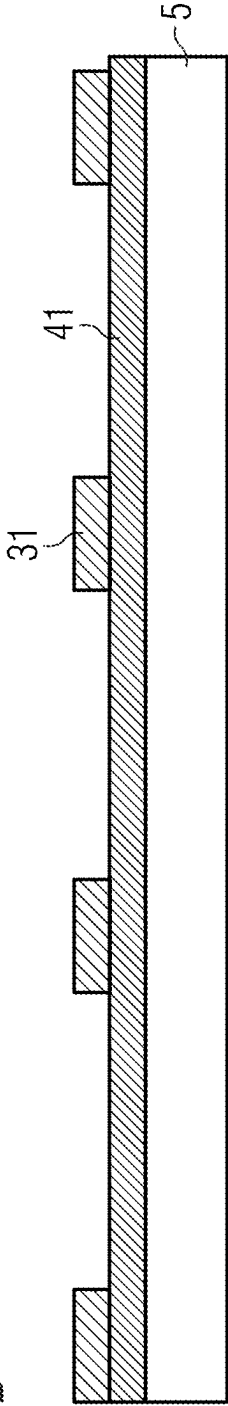


FIG 1C1

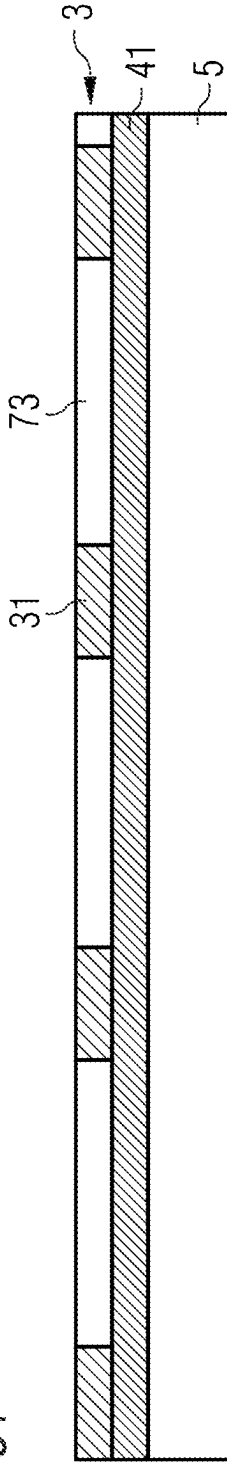


FIG 1C2

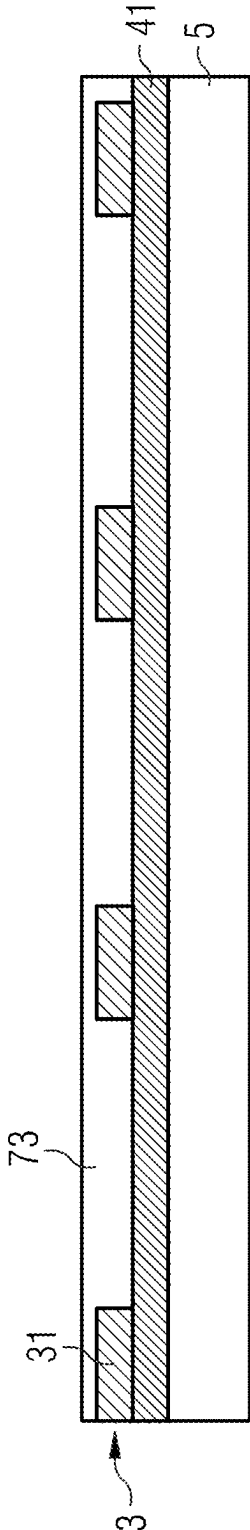


FIG 1D

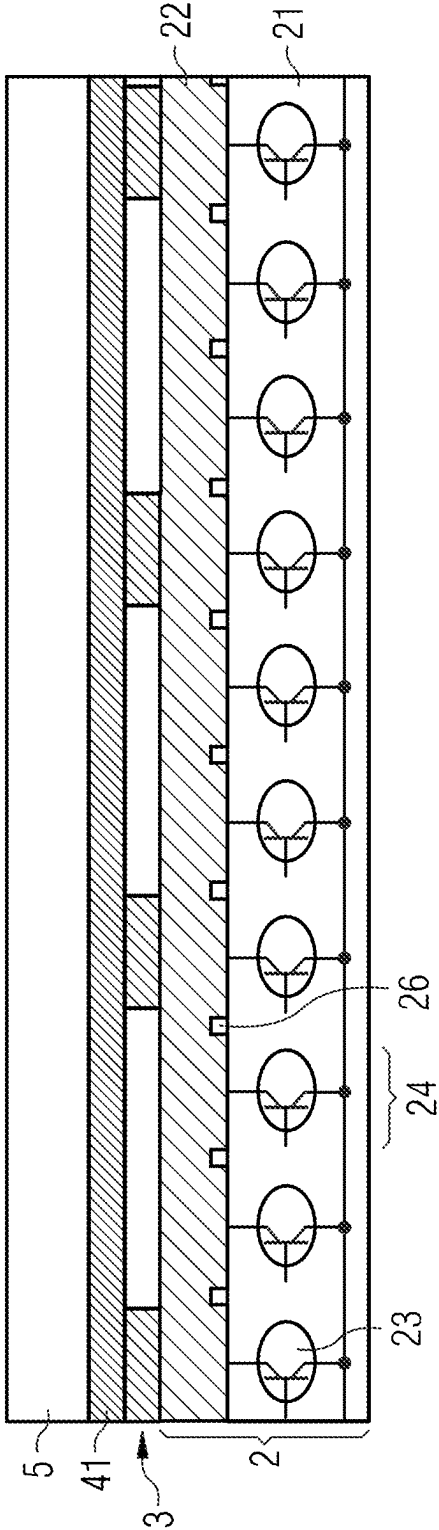


FIG 1E

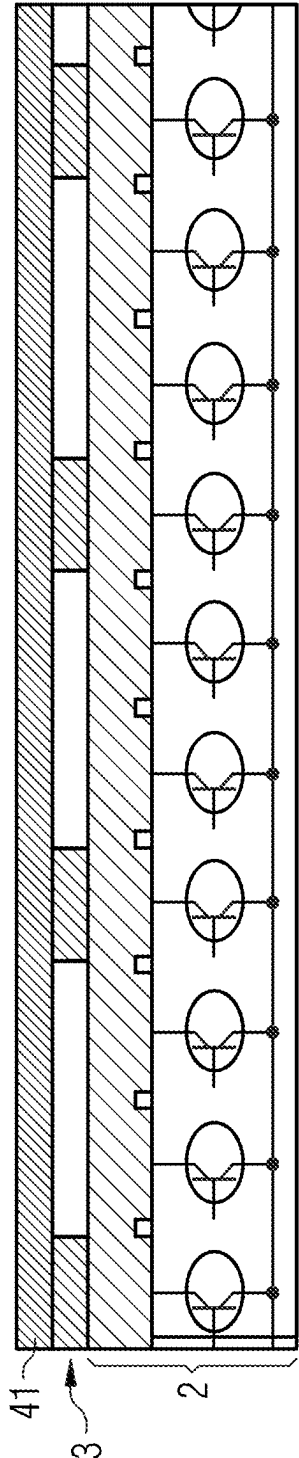


FIG 1F

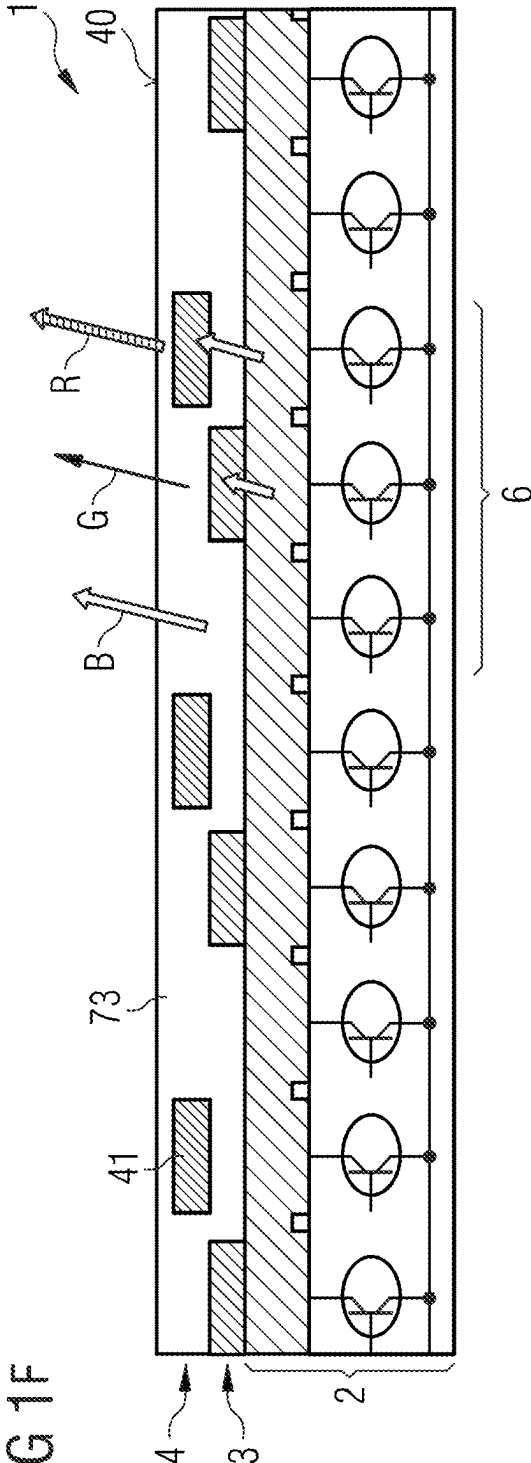


FIG 2A

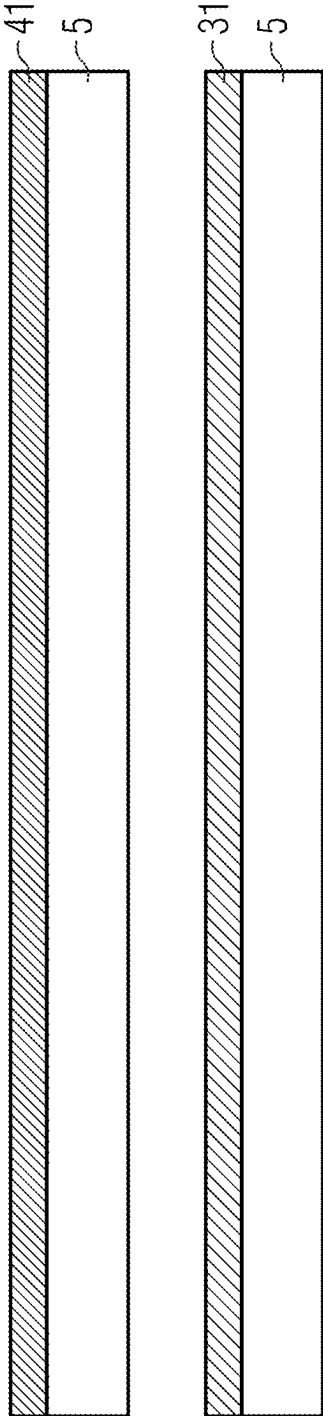


FIG 2B

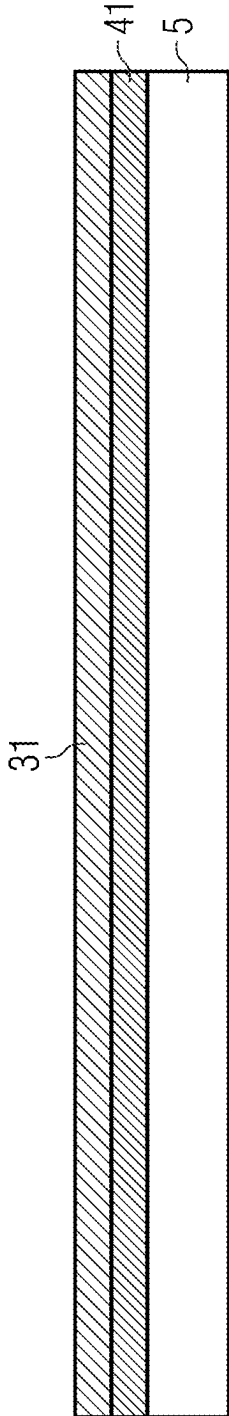


FIG 3A

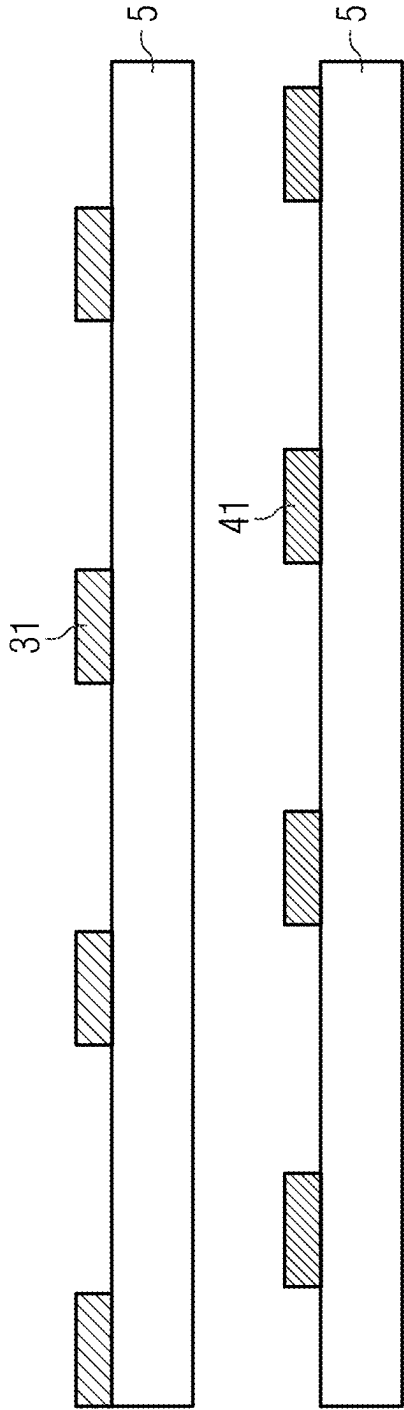


FIG 3B

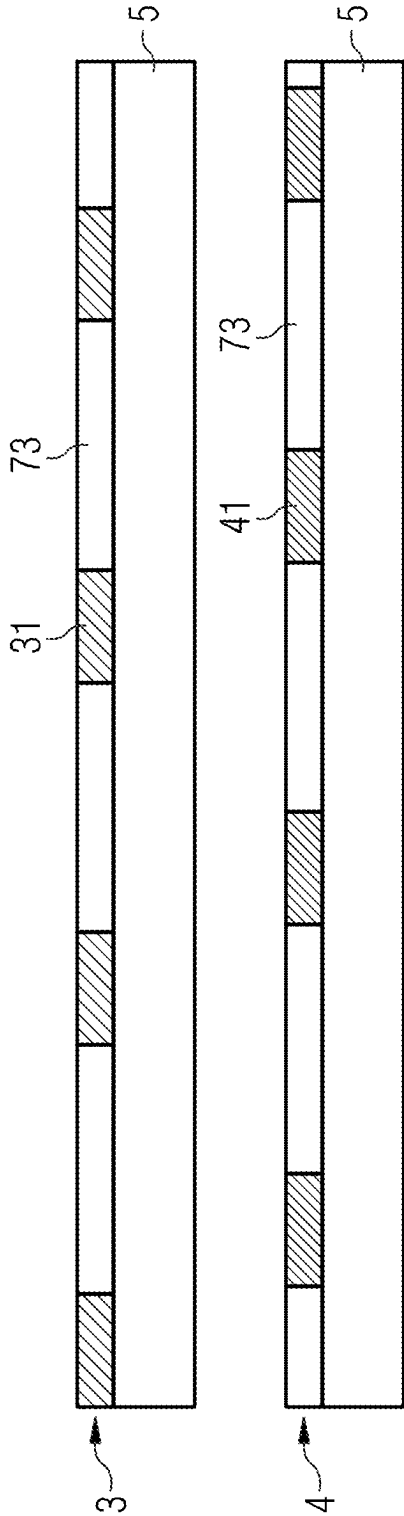


FIG 3C

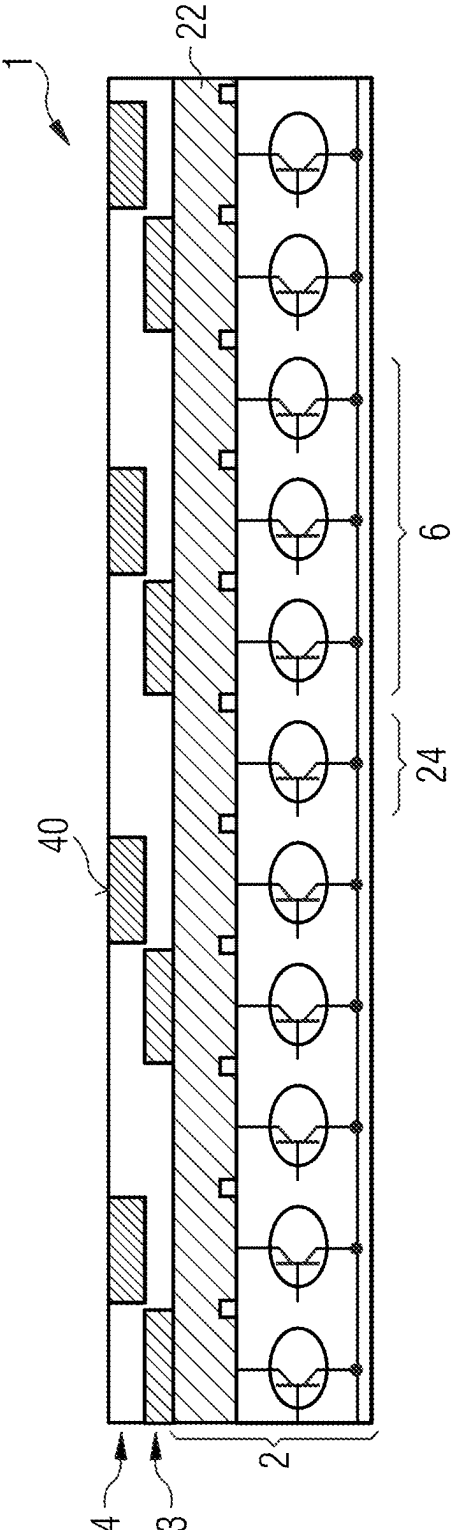


FIG 4

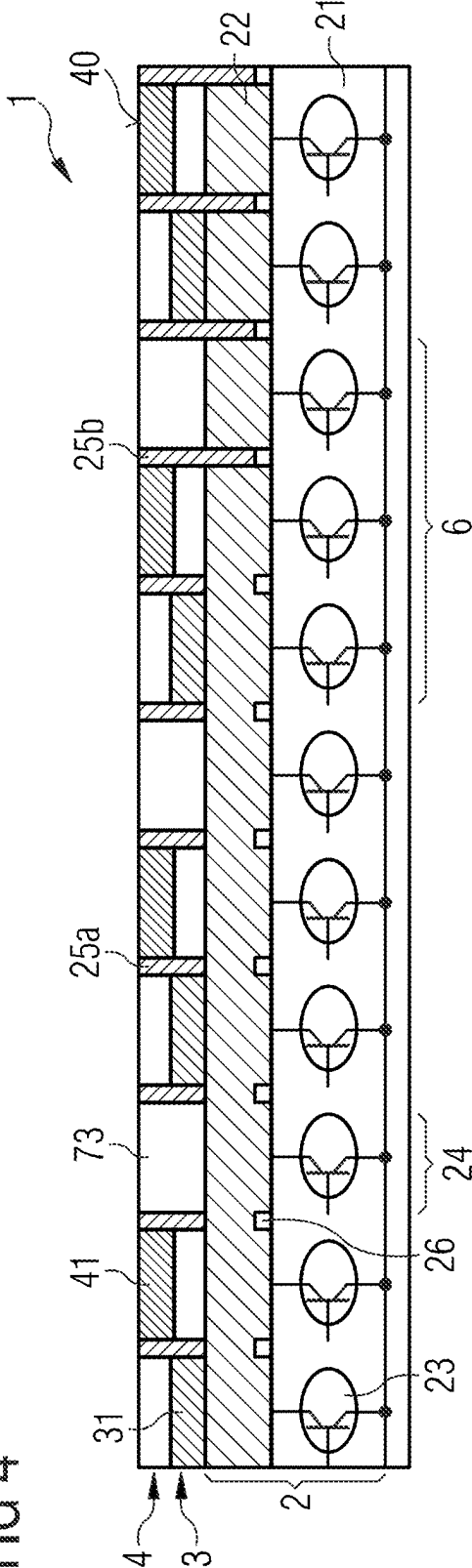


FIG 5

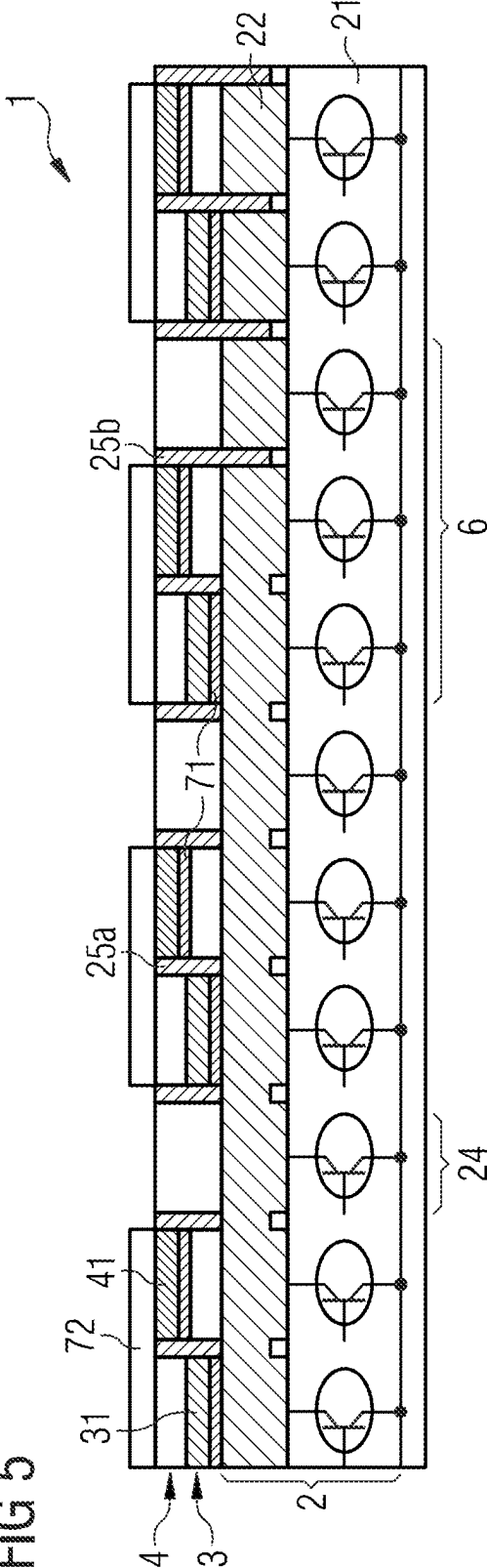




FIG 6 A

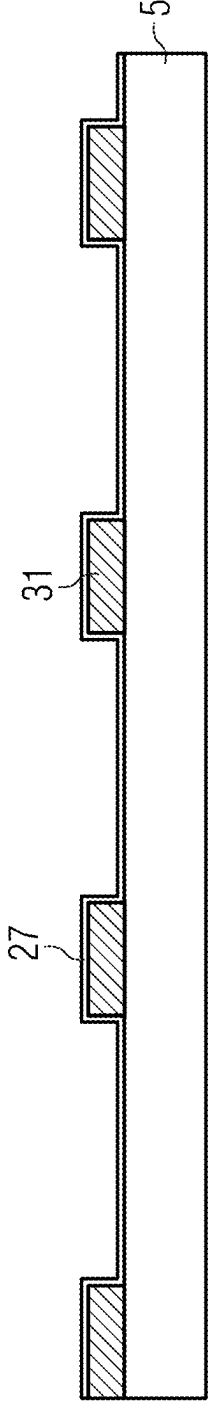
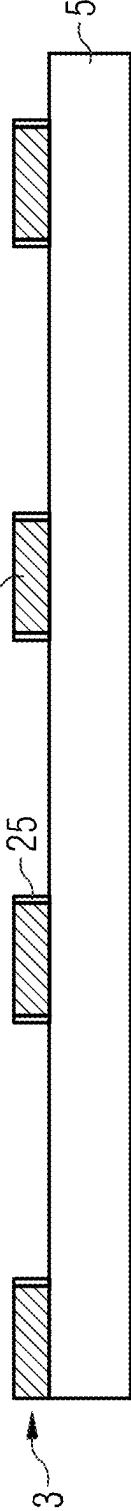


FIG 6 B





**METHOD OF PRODUCING  
OPTOELECTRONIC SEMICONDUCTOR  
COMPONENTS AND AN OPTOELECTRONIC  
SEMICONDUCTOR COMPONENT**

**CROSS-REFERENCE TO RELATED  
APPLICATIONS**

**[0001]** This application is a divisional application of U.S. application Ser. No. 18/082,440, filed Dec. 15, 2022, which is a continuation of U.S. application Ser. No. 17/206,336, filed Mar. 19, 2021 (now U.S. Pat. No. 11,557,700), which is a continuation of U.S. application Ser. No. 16/335,632, filed Mar. 21, 2019 (now U.S. Pat. No. 11,018,283), which is a 35 U.S.C. § 371 of International Application No. PCT/EP2017/077326, filed Oct. 25, 2017, which claims priority to German Patent Application No. 102016220915.9, filed Oct. 25, 2016, each of which is incorporated herein by reference in its entirety.

**TECHNICAL FIELD**

**[0002]** This disclosure relates to a method of producing optoelectronic semiconductor components and an optoelectronic semiconductor component.

**BACKGROUND**

**[0003]** There is a need to provide an optoelectronic semiconductor component with many pixels, that is adjustable to emit light in different colors, and can be produced efficiently.

**SUMMARY**

**[0004]** We provide a method of producing optoelectronic semiconductor components including providing a primary light source having a carrier and a semiconductor layer sequence mounted thereon that generates primary light (B), wherein the semiconductor layer sequence is structured into a plurality of pixels that can be driven electrically independently of each other, and the carrier includes a plurality of control units that drive the pixels, providing at least one conversion unit adapted to convert the primary light (B) into at least one secondary light (G, R), wherein the conversion unit is grown continuously from at least one semiconductor material, structuring the conversion unit, wherein portions of the semiconductor material are removed in accordance with the pixels, and applying the conversion unit to the semiconductor layer sequence so that the remaining semiconductor material is uniquely assigned to a portion of the pixels.

**[0005]** We also provide an optoelectronic semiconductor component including a primary light source including a carrier and a semiconductor layer sequence mounted thereon that generate primary light (B), and at least one conversion unit of at least one semiconductor material adapted to convert the primary light (B) into at least one secondary light (G, R), wherein the semiconductor layer sequence and the conversion unit are produced separately from each other and are not grown together, the semiconductor layer sequence is structured into a plurality of pixels that can be controlled electrically independently of each other, the carrier includes a plurality of control units that drive the pixels, no conversion unit is assigned to some of the pixels so that selected pixels emit the primary light (B), and exactly one conversion unit each is assigned to remaining pixels, a plurality of pixels emitting different colors are combined to form a display area that is adjustable to emit light of different

colors, and a light path between the carrier and a light-exit side of the conversion unit facing away from the carrier is free of organic materials.

**[0006]** We further provide an optoelectronic semiconductor component including a primary light source including a carrier and a semiconductor layer sequence mounted thereon and configured to generate primary light, and at least one conversion unit of at least one semiconductor material adapted to convert the primary light into at least one secondary light, wherein the semiconductor layer sequence and the converter unit are separate elements, the semiconductor layer sequence includes a plurality of pixels, the pixels are configured to be controlled electrically independently of each other, the carrier includes a plurality of control units configured to drive the pixels, all pixels of a first group are free of a conversion unit, and the pixels of the first group are configured to emit the primary light, all pixels of a second group of pixels include exactly one conversion unit each, and the pixels of the second group are configured to emit the at least one secondary light, a plurality of pixels from the first group and the second group are combined to form a display area that is adjustable to emit light of different colors, and a light path between the carrier and a light-exit side of the conversion unit facing away from the carrier is free of organic materials.

**[0007]** We still further provide an optoelectronic component including a primary light source including a carrier and a semiconductor layer sequence mounted thereon that generate primary light, and at least one conversion unit of at least one semiconductor material adapted to convert the primary light into at least one secondary light, wherein the semiconductor layer sequence and the conversion unit are produced separately from each other and are not grown together, the semiconductor layer sequence is structured into a plurality of pixels that can be controlled electrically independently of each other, the carrier includes a plurality of control units that drive the pixels, no conversion unit is assigned to some of the pixels so that selected pixels emit the primary light, and exactly one conversion unit each is assigned to remaining pixels, a plurality of pixels emitting different colors are combined to form a display area that is adjustable to emit light of different colors, and a light path between the carrier and a light-exit side of the conversion unit facing away from the carrier is free of organic materials.

**[0008]** We yet further provide an optoelectronic semiconductor component including a primary light source including a carrier and a semiconductor layer sequence mounted thereon and configured to generate primary light, and at least one conversion unit of at least one semiconductor material adapted to convert the primary light into at least one secondary light, wherein the semiconductor layer sequence and the converter unit are separate elements, the semiconductor layer sequence includes a plurality of pixels, the pixels are configured to be controlled electrically independently of each other, the carrier includes a plurality of control units configured to drive the pixels, all pixels of a first group are free of a conversion unit, and the pixels of the first group are configured to emit the primary light, all pixels of a second group of pixels include exactly one conversion unit each, and the pixels of the second group are configured to emit the at least one secondary light, a plurality of pixels from the first group and the second group are combined to

form a display area that is adjustable to emit light of different colors, and adjacent pixels are optically isolated from each other.

**[0009]** We still yet further provide an optoelectronic semiconductor component including a primary light source including a carrier and a semiconductor layer sequence mounted thereon and configured to generate primary light, and at least one conversion unit of at least one semiconductor material adapted to convert the primary light into at least one secondary light, wherein the semiconductor layer sequence and the converter unit are separate elements, the semiconductor layer sequence includes a plurality of pixels, the pixels are configured to be controlled electrically independently of each other, the carrier includes a plurality of control units configured to drive the pixels, all pixels of a first group are free of a conversion unit, and the pixels of the first group are configured to emit the primary light, all pixels of a second group of pixels include exactly one conversion unit each, and the pixels of the second group are configured to emit the at least one secondary light, a plurality of pixels from the first group and the second group are combined to form a display area that is adjustable to emit light of different colors, and the conversion unit is attached to the semiconductor layer sequence by direct bonding or anodic bonding or adhesion bonding.

**[0010]** We yet further provide a method of producing optoelectronic semiconductor components including providing a primary light source having a carrier and a semiconductor layer sequence mounted thereon that generates primary light (B), wherein the semiconductor layer sequence is structured into a plurality of pixels that can be driven electrically independently of each other, and the carrier includes a plurality of control units that drive the pixels, providing at least one conversion unit adapted to convert the primary light (B) into at least one secondary light (G, R), wherein the conversion unit is grown continuously from at least one semiconductor material, structuring the conversion unit, wherein portions of the semiconductor material are removed in accordance with the pixels, and applying the conversion unit to the semiconductor layer sequence so that the remaining semiconductor material is uniquely assigned to a portion of the pixels, wherein at least two conversion units are grown on a common growth substrate, and at least one of the conversion units is structured while still applied to the common growth substrate.

#### BRIEF DESCRIPTION OF THE DRAWINGS

**[0011]** FIGS. 1A-1F, 2A-2B, 3A-3C, and 6A-6B are schematic sectional representations of process steps of examples of methods described for the production of optoelectronic semiconductor components.

**[0012]** FIGS. 4, 5 and 7 are schematic sectional views of examples of optoelectronic semiconductor components.

#### REFERENCE CHARACTER LIST

- [0013]** 1 optoelectronic semiconductor component
- [0014]** 2 primary light source
- [0015]** 21 carrier
- [0016]** 22 semiconductor layer sequence
- [0017]** 22a first semiconductor layer sequence
- [0018]** 22b second semiconductor layer sequence
- [0019]** 23 control unit
- [0020]** 24 pixels

- [0021]** 25 opaque partition wall
- [0022]** 26 separation region
- [0023]** 27 raw material layer
- [0024]** 3 first conversion unit
- [0025]** 31 semiconductor material
- [0026]** 4 second conversion unit
- [0027]** 40 light-exit side
- [0028]** 41 semiconductor material
- [0029]** 5 growth substrate
- [0030]** 6 display area (pixels)
- [0031]** 71 mirror layer
- [0032]** 72 filter layer
- [0033]** 73 planarization layer
- [0034]** B primary light
- [0035]** B1 first primary light
- [0036]** G first secondary light
- [0037]** G2 second primary light
- [0038]** R second secondary light
- [0039]** R1 secondary light

#### DETAILED DESCRIPTION

**[0040]** Our method produces optoelectronic semiconductor components. Several of the semiconductor components can be produced together in a wafer composite. The optoelectronic semiconductor components are adapted to generate color-adjustable light. The produced semiconductor components serve, for example, as display devices, as displays or in spotlights with adjustable emission characteristics, for example, in motor vehicles.

**[0041]** The method may comprise the step of providing a primary light source. The primary light source generates electromagnetic radiation, especially primary light, via electroluminescence. Primary light is preferably blue light, for example, with a wavelength of maximum intensity of at least 420 nm or 435 nm and/or at most 480 nm or 460 nm.

**[0042]** The primary light source may alternatively or additionally generate electromagnetic radiation in other wavelength ranges. For example, the primary light source may produce ultraviolet radiation alternatively or in addition to blue light, for example, with a wavelength of maximum intensity at 365 nm minimum and/or 420 nm maximum. For example, the primary light source may produce green light alternatively or in addition to blue light, for example, with a wavelength of maximum intensity at 485 nm minimum and/or 575 nm maximum.

**[0043]** The primary light source may comprise a carrier. The carrier contains a large number of control units. In particular, the carrier is a silicon-based semiconductor carrier comprising transistors and/or switching units and/or control units. The control units can be produced in CMOS technology. The carrier is preferably based on a semiconductor material such as silicon or germanium.

**[0044]** A semiconductor layer sequence may be attached to the carrier. The semiconductor layer sequence is adapted to generate the primary radiation. For this purpose, the semiconductor layer sequence comprises at least one active zone. The semiconductor layer sequence is preferably based on a III-V compound semiconductor material. For example, the semiconductor material is a nitride compound semiconductor material such as AlInIn1-n-mGamN or a phosphide compound semiconductor material such as AlInIn1-n-mGamP or also an arsenide compound semiconductor material such as AlInIn1-n-mGamAs or such as AlInIn1-n-mGamAskP1-k, wherein:  $0 \leq n \leq 1$ ,  $0 \leq m \leq 1$  and  $n+m \leq 1$  and  $0 \leq k < 1$ .

Preferably,  $0 < n \leq 0.8$ ,  $0.4 \leq m < 1$  and  $n + m \leq 0.95$  and  $0 < k \leq 0.5$  applies to at least one layer or to all layers of the semiconductor layer sequence. The semiconductor layer sequence may contain dopants and additional components. For the sake of simplicity, however, only the essential components of the crystal lattice of the semiconductor layer sequence are indicated, i.e., Al, As, Ga, In, N or P, even if they may be partially replaced and/or supplemented by small quantities of other substances.

**[0045]** In the method, the semiconductor layer sequence may be structured into a plurality of electrically independently controllable pixels in plan view. This structuring is carried out in particular by removing material from part of the semiconductor layer sequence. In other words, the semiconductor layer sequence is structured by etching. Preferably, material is removed from the semiconductor layer sequence between adjacent pixels.

**[0046]** The pixels may be electrically contacted independently of each other and can be controlled independently of each other. The independent control is effected via the control units in the carrier. Preferably, one of the control units of the carrier is assigned to each of the pixels and vice versa.

**[0047]** The method may include the step of providing one or more conversion units. At least one conversion unit is adapted to partially or completely absorb the primary light and convert it into at least one secondary light. The secondary light is generated by photoluminescence from the primary light.

**[0048]** The conversion unit or all conversion units may be grown contiguously from at least one semiconductor material. In particular, each conversion unit is epitaxially grown. The conversion units can be based on the same semiconductor materials as the semiconductor layer sequence of the primary light source, or on different semiconductor materials.

**[0049]** The conversion units that generate the secondary light may each comprise a single quantum well structure or, preferably, a multiple quantum well structure. This means that the conversion units can be composed of a sequence of several semiconductor layers. In a lateral direction, in particular parallel to a main expansion direction of the carrier, a composition of the conversion units prefers not to change or not to change significantly. In other words, the semiconductor layers of the conversion units continue over the entire respective conversion unit, preferably in an unchanged composition within the manufacturing tolerances.

**[0050]** The conversion units may be structured. During structuring, the semiconductor material from which the conversion units are made is removed in subareas. It is preferred to completely remove the semiconductor material in the sub-areas so that, seen from above, recesses and/or holes are created in the conversion units. Furthermore, it is possible that only islands of the semiconductor material of the conversion units remain, which are not interconnected by semiconductor material of the respective conversion unit.

**[0051]** Structuring of the at least one conversion unit may take place according to the pixels of the semiconductor layer sequence. For example, the same grid is used to structure the semiconductor material used to structure the semiconductor layer sequence into the pixels. Thus, it is possible to assign at least a part of the pixels to each area of the semiconductor material of the conversion unit so that the corresponding pixel generates the secondary light during operation.

**[0052]** The structured conversion unit may be applied to the semiconductor layer sequence. This results in an assignment of the remaining semiconductor material of the conversion layers after structuring to a part of the pixels. This assignment is preferably unambiguous.

**[0053]** The method may be established to produce optoelectronic semiconductor components and may comprise the following steps, for example, in the given order:

**[0054]** providing a primary light source having a carrier and a semiconductor layer sequence attached to generate primary light, wherein the semiconductor layer sequence in plan view is structured into a plurality of electrically independently controllable pixels, and the carrier comprises a plurality of driving units that control the pixels,

**[0055]** providing at least one conversion unit adapted to convert the primary light into at least one secondary light, wherein the conversion unit is grown continuously from at least one semiconductor material,

**[0056]** structuring the conversion unit, wherein partial regions of the semiconductor material are removed according to the pixels, and

**[0057]** applying the conversion unit to the semiconductor layer sequence so that the remaining semiconductor material is uniquely assigned to a part of the pixels.

**[0058]** The primary light source may be constructed as specified in DE 10 2014 101 896 A1 or DE 10 2014 105 999 A1, the subject matter of which with regard to the primary light source is incorporated herein by reference. Such primary light sources comprising a carrier with control units and a semiconductor layer sequence structured in pixels are also referred to as microlight emitting diodes, or micro LEDs for short.

**[0059]** Micro LEDs are mostly monochromatic emitting components. To produce a colored image with three monochromatic micro LEDs, one for red light, one for green light and one for blue light, the emitted light by the micro LEDs can be mixed and superimposed via a prism. It is also possible to generate a color image directly from the emission of individual pixels, wherein individual different colors emitting pixels of a semiconductor material can either be placed side by side or stacked vertically on top of each other. However, such arrangements are either very time-consuming to produce due to the fact that individual pixels have to be placed individually, or have additional optical components.

**[0060]** With our method, on the other hand, a semiconductor component in the form of a micro LED can be efficiently produced, wherein the individual pixels result from a single semiconductor layer sequence. By applying at least one conversion unit, compact and efficiently producible pixels can be generated that emit light in different colors. This is achieved in particular by structuring the conversion units according to the division of the semiconductor layer sequence into pixels. Both the conversion units and the semiconductor layer sequence can be produced at wafer level.

**[0061]** By generating different colors by photoluminescence in the conversion units, it is possible to use a monochromatically emitting primary light source in the form of a micro LED. For example, the arrangement of electrical contacts and passivation layers is comparatively simple and efficient compared to other methods. The use of epitaxially grown semiconductor layers for the conversion units enables

generation of a layer stack containing several layers emitting in different wavelengths. Furthermore, with such semiconductor layers for the conversion units it is possible to achieve a high degree of absorption, especially for blue light so that a local full conversion of the primary light can be achieved. In addition, spectrally narrow-band spectra of the respective photoluminescence radiation can be obtained. This opens up a wide gamut range in display applications, for example. Since at a certain location above the semiconductor layer sequence only one single conversion unit is particularly preferred, undesired absorption losses can be avoided. The use of planarization processes to fill gaps between semiconductor material areas of the conversion units enables mechanical stabilization. In addition, the conversion units can be directly bonded to the primary light source.

**[0062]** When structuring the semiconductor layer sequence to the pixels, there need not be a change in the position of remaining regions of the semiconductor layer sequence relative to each other. In other words, the areas for the individual pixels are not first juxtaposed by a subsequent placement process, but the pixels are created directly from the semiconductor layer sequence without subsequent rearrangement or repositioning. This also applies to the structuring of the conversion units. This means that even the remaining parts of the semiconductor material of the conversion units are not changed with regard to their position and position relative to each other after or during structuring.

**[0063]** Several conversion units may be applied to the semiconductor layer sequence. Exactly two conversion units are preferred. A first conversion unit converts blue light into green light, and a second conversion unit converts blue light into red light. If the primary light source emits ultraviolet radiation and little or no blue light, a third conversion unit is preferred to generate blue light.

**[0064]** The semiconductor layers of the conversion units may be grown on top of each other on a common growth substrate. In other words, the conversion units can form a common, coherently grown semiconductor layer sequence.

**[0065]** One or more of the conversion units may be structured on the common growth substrate. The structuring is, for example, photolithographically done and by etching. It is preferable to structure the conversion units one after the other in time.

**[0066]** In the method, only one of the conversion units need be structured on the common growth substrate. After structuring this conversion unit, the compound of the conversion units and the common growth substrate may be attached to the semiconductor layer sequence, preferably by wafer bonding. The growth substrate is then removed, for example, by laser treatment, etching and/or mechanical methods such as grinding. Finally, at least one further conversion unit is structured, wherein this structuring takes place at the semiconductor layer sequence of the primary light source.

**[0067]** Several conversion units may be applied to the semiconductor layer sequence of the primary light source. Each of the conversion units is grown on its own growth substrate. The conversion units thus each represent a separately produced, separate semiconductor layer sequence.

**[0068]** The conversion units may be structured on the respective growth substrate. This makes it possible to avoid structuring the conversion units directly on the semiconductor layer sequence.

**[0069]** One or more planarization layers may be applied after structuring the corresponding conversion unit. The planarization layer is made of a radiation-permeable material and preferably consists of inorganic materials, for example, an oxide such as silicon oxide or aluminum oxide or a nitride such as silicon nitride or aluminum nitride.

**[0070]** The planarization layer may mechanically connect individual island-like areas of the conversion units resulting from the structuring of the semiconductor material. A material for the planarization layer can be located directly on the semiconductor material of the conversion units. Alternatively, it is possible that a coating, especially an optically functionalized coating such as an optical insulation or a mirror is located between the material of the planarization layer and the semiconductor material of the conversion units.

**[0071]** The planarization layer may partially or completely cover a side of the conversion unit facing away from the associated growth substrate. The planarization layer can cover this side permanently or only temporarily during certain producing steps.

**[0072]** The conversion units may each be grown on separate growth substrates. Subsequently, the conversion units may be attached to each other before the growth substrates are detached, in particular by direct bonding with or without an intermediate layer. This creates a layer sequence from the conversion units.

**[0073]** The sequence of layers from the conversion units may be structured, where in particular, only one of the conversion units is structured. Subsequently, this layer sequence is attached to the semiconductor layer sequence of the primary light source. The remaining growth substrate can be removed then. The previously unstructured conversion unit can then be structured.

**[0074]** The conversion units may be attached to each other or the conversion units may be attached to the semiconductor layer sequence by wafer bonding. This can be wafer bonding without an intermediate layer, in particular so-called direct bonding or anodic bonding. Alternatively, a particularly oxidic or nitridic intermediate layer can be used and wafer bonding is eutectic bonding, glass bonding or adhesive bonding.

**[0075]** A light path between the carrier and a light exit side of the conversion units facing away from the carrier may be free of organic material. In other words, light may only pass through inorganic materials within the semiconductor component. This results in a long service life and high power densities, especially of blue light, can be achieved.

**[0076]** Some of the pixels may not be assigned to any of the conversion units. The primary light is emitted from these pixels without wavelength conversion. Furthermore, exactly one of the conversion units may be assigned to each of the other pixels. This means that there are no conversion units stacked on top of each other at the pixels. For example, pixels that emit the primary light without wavelength conversion are present that emit blue light. In addition, pixels may be present that emit the primary light without wavelength conversion emitting green light. For example, it is possible that blue and green light are generated by an electrically pumped stack of layers and red light is generated by an optically pumped semiconductor layer.

**[0077]** Several, in particular exactly three, differently colored emitting pixels may be combined to one display area. Such display areas are also referred to as colored pixels. The

display areas can emit light in different colors. By arranging the pixels in display areas, it is possible to use the semiconductor component to display images or films or varying light patterns.

**[0078]** Adjacent pixels and/or adjacent display areas may be optically isolated from each other by a partition wall. The partition wall can be composed of one or more opaque materials. The partition may be absorbent or reflective to the radiation produced.

**[0079]** The partition wall may extend completely through the conversion unit. It is possible for the partition wall to be flush with the semiconductor layer sequence or partially or completely penetrate the semiconductor layer sequence. If the intermediate wall protrudes into the semiconductor layer sequence, the intermediate wall is, for example, made of an electrically insulating material or electrically insulating towards the semiconductor layer sequence. Alternatively, it is possible that, for example, an n-conducting side of the semiconductor layer sequence is electrically contacted via the partition walls so that a common electrical contact across the pixels can be realized through the partition wall.

**[0080]** The semiconductor layer sequence may be continuous and contiguous and preferably extends seamlessly across all pixels. Alternatively, the semiconductor layer sequence may be completely removed in areas between adjacent pixels and/or display areas so that the pixels or display areas are realized by islands out of the semiconductor layer sequence.

**[0081]** At least one mirror layer may be applied to one side of the conversion units facing the semiconductor layer sequence. The one or more mirror layers are impermeable or largely impermeable to the secondary light generated in the associated conversion unit. The mirror layers are preferably used to reflect this secondary light. In particular, the mirror layers are transparent to the primary light and realized by dichroic mirrors.

**[0082]** There may be at least one filter layer on one side of the conversion units facing away from the carrier. The filter layer is impermeable to primary light and permeable to secondary light. The filter layer can absorb or reflect the primary light. Optionally, the filter layer can be designed as an anti-reflective layer for the secondary light.

**[0083]** The filter layer and/or mirror layer may completely cover the associated conversion unit. Alternatively, the associated conversion unit is only partially covered by the filter layer and/or mirror layer.

**[0084]** At least one of the conversion units or all conversion units and/or the semiconductor layer sequence may have a thickness of at least 1  $\mu\text{m}$  or 2  $\mu\text{m}$  and/or a thickness of at most 15  $\mu\text{m}$  or 10  $\mu\text{m}$  or 6  $\mu\text{m}$ . In other words, the conversion units and the semiconductor layer sequence are thin.

**[0085]** The pixels may have an average diameter of at least 2  $\mu\text{m}$  or 3  $\mu\text{m}$  or 10  $\mu\text{m}$  in plan view. Alternatively or additionally, the mean diameter is at most 300  $\mu\text{m}$  or 200  $\mu\text{m}$  or 80  $\mu\text{m}$ . In particular, these values apply not only to the mean diameter, but also to a mean edge length of the pixels if they are square or rectangular in shape in plan view.

**[0086]** A distance between adjacent pixels may be at least 0.3  $\mu\text{m}$  or 0.5  $\mu\text{m}$  or 1  $\mu\text{m}$  and/or at most 10  $\mu\text{m}$  or 6  $\mu\text{m}$  or 3  $\mu\text{m}$ . The distance between the pixels shall preferably not exceed 25% or 10% or 3% of the mean diameter of the pixels.

**[0087]** The finished semiconductor device may comprise at least 10 or 100 or 1000 of the pixels. Alternatively or additionally, the number of pixels shall not exceed 108 or 107 or 106 or 105. The same may apply to the number of display areas.

**[0088]** We also provide an optoelectronic semiconductor device. The semiconductor device is preferably produced by methods as indicated in conjunction with one or more of the above examples. Features for the methods are therefore also disclosed for the semiconductor device and vice versa.

**[0089]** The optoelectronic semiconductor device may comprise a primary light source comprising a carrier and a semiconductor layer sequence attached to generate primary light. The semiconductor device further comprises at least one conversion unit of at least one semiconductor material, the conversion unit being adapted to convert the primary light into at least one secondary light by photoluminescence. The semiconductor layer sequence and the conversion unit are produced separately from each other and are not grown together. Seen from above, the semiconductor layer sequence is structured into a large number of electrically independently controllable pixels. The carrier comprises a large number of control units that control the pixels and are preferably available in a 1:1 assignment to the pixels. Some of the pixels are not assigned a conversion unit so that these pixels emit the primary light and the remaining pixels are each assigned exactly one conversion unit.

**[0090]** Several pixels emitting different colors are combined to form a display area designed to emit light that can be adjusted in terms of color. Furthermore, a light path between the carrier and a light exit side of the conversion unit facing away from the carrier is free of organic materials.

**[0091]** In the following, a process described here and a semiconductor component described here are explained in more detail with reference to the drawing using examples. The same reference signs indicate the same elements in the individual figures. However, no true-to-scale relations are shown. Rather, individual elements may be exaggeratedly large for a better understanding.

**[0092]** FIGS. 1A-1F schematically illustrate a method of producing optoelectronic semiconductor components 1. According to FIG. 1A, a growth substrate 5 is provided. Growth substrate 5, for example, is a GaAs substrate. Two semiconductor materials 31, 41 are grown on growth substrate 5 for two conversion units 3, 4. First, a second semiconductor material 41 is grown to generate red light and then a first semiconductor material 31 to generate green light.

**[0093]** The two semiconductor materials 31, 41 are based on InGaAlP. The second semiconductor material 41 that generates red light is preferably based on InGaAlP. As an alternative to InGaAlP, the first semiconductor material 31 can also be based on InGaN, grown, for example, on a sapphire substrate. Such a translucent sapphire substrate can also remain attached to the semiconductor material 31, in contrast to FIGS. 1A-1F. The semiconductor materials 31, 41 each consist of several layers and preferably comprise a multi-quantum well structure adapted to generate green or red secondary light by photoluminescence.

**[0094]** For example, a multi-quantum well structure may comprise ten or more quantum well structures. For example, the multi-quantum well structure in the first semiconductor material 31 may comprise twenty or more quantum well structures. Furthermore, the multi-quantum well structure in

the first semiconductor material **31** may comprise, for example, one hundred or fewer quantum well structures. For example, the multi-quantum well structure in the second semiconductor material **41** may comprise twenty or more quantum well structures. Furthermore, the multi-quantum well structure in the second semiconductor material **41** may, for example, comprise one hundred or fewer quantum well structures. According to one example, the number of quantum well structures in the first and/or second semiconductor material is between twenty and fifty quantum well structures. The number of quantum well structures in the first semiconductor material **31** and the number of quantum well structures in the second semiconductor material **41** may be different. Further, the number of quantum well structures in the first and/or second semiconductor material **31**, **41** may be greater than the number of quantum well structures in the semiconductor layer sequence **22** of the primary light source **2**.

**[0095]** The process step of FIG. 1B shows that the first semiconductor material **31** is structured. The first semiconductor material **31** is completely removed in areas, in particular by etching so that the second semiconductor material **41** is exposed in areas. Otherwise, the second semiconductor material **41** remains unaffected by this structuring.

**[0096]** The structuring of the first semiconductor material **31** is carried out, for example, by wet chemical or dry chemical etching. The individual, remaining island-shaped regions of the first semiconductor material **31** correspond to the size of pixels **24** of the semiconductor component **1**. The island-shaped regions have, for example, an edge length of at least 3  $\mu\text{m}$  and/or at most 200  $\mu\text{m}$ . Seen from above, these areas are square, rectangular, round or hexagonal. These areas can be cartesian in a rectangular pattern or hexagonal. The same applies to the second semiconductor material **41** and also to all other examples.

**[0097]** In the process step of FIG. 1C1, a planarization layer **73** is deposited over the entire surface. The planarization layer **73** is preferably made of an electrically insulating dielectric material such as silicon dioxide. The planarization layer **73** fills gaps between the island-like, remaining regions of the first semiconductor material **31** so that a total layer with a constant thickness is formed. In addition, the planarization layer **73** firmly connects the island-shaped regions of the first semiconductor material **31** with each other. The planarization layer **73** forms together with the first semiconductor material **31** the first conversion unit **3**.

**[0098]** According to FIG. 1C1, the planarization layer **73** has the same thickness as the first semiconductor material **31** so that the first semiconductor material **31** and the planarization layer **73** are flush with each other, directing away from the growth substrate **5**. In contrast, FIG. 1C2 illustrates that a thickness of the planarization layer **73** is greater than a thickness of the first semiconductor material **31**. On a side facing away from the growth substrate **5**, the first semiconductor material **31** is completely covered by the planarization layer **73**. The planarization layer **73** together with the first semiconductor material **31** again has a constant thickness. This thickness is at least 5  $\mu\text{m}$  and/or at most 12  $\mu\text{m}$ , for example, with the planarization layer **73** preferably exceeding the first semiconductor material **31** by at most 5  $\mu\text{m}$  or 2  $\mu\text{m}$  or 1  $\mu\text{m}$ . This also applies preferentially to all other examples.

**[0099]** The variants with regard to the planarization layer **73**, as illustrated in FIGS. 1C1 and 1C2, can also be found in all other examples. To simplify the illustration, only one of these variants is illustrated below.

**[0100]** FIG. 1D shows that the composite of FIG. 1C1, or alternatively FIG. 1C2, is attached to a primary light source **2**. This is preferably done by direct bonding. The composite of FIG. 1C1 can directly connect to a semiconductor layer sequence **22** of the primary light source **2** as shown in FIG. 1D. Alternatively, it is possible to apply an unrepresented intermediate layer such as silicon dioxide, between the primary light source **2** and the composite of FIG. 1C1 to mediate adhesion. Alternatively, the planarization layer **73** from FIG. 1C2 can be used as an adhesion promoter.

**[0101]** The primary light source **2** also has a carrier **21**. The carrier **21** contains a large number of control units **23**. The carrier **21** is preferably based on silicon, and the control units **23** are produced in CMOS technology in the carrier **21**. The semiconductor layer sequence **22** is based on AlInGaN and is designed to generate blue light. The semiconductor layer sequence **22** is divided into a number of pixels **24**. Each of the pixels **24** is preferably assigned to exactly one of the control units **23** and vice versa.

**[0102]** Separation areas **26** are optionally located between adjacent pixels **24**. Electrical and/or optical isolation of the individual pixels **24** from each other can be achieved via the separation areas **26**. The semiconductor layer sequence **22** is only partially interrupted by the separation areas **26**, for example, in form of unfilled or filled trenches, and extends as a continuous layer over the entire carrier **21**. Seen from above, the pixels **24** are, for example, rectangular, square, round or hexagonal.

**[0103]** At the process step of FIG. 1E, the growth substrate **5** from FIG. 1D is removed. The growth substrate **5** is removed by grinding, polishing, wet or dry etching, laser lifting or combinations thereof.

**[0104]** Subsequently, in FIG. 1F, the second semiconductor material **41** is structured in the same way as the first semiconductor material **31**, wherein the structuring is performed on the semiconductor layer sequence **22**. The semiconductor layer sequence **22** is not affected by this structuring. A further planarization layer **73** is also applied. The further planarization layer **73** forms together with the second semiconductor material **41** the second conversion unit **4**.

**[0105]** The island-shaped remaining areas of the two semiconductor materials **31**, **41** have the same size as the areas of the semiconductor layer sequence **22** for the pixels **24**. For example, a third of the pixels **24** is free for the conversion units **3**, **4** in an RGB unit or a quarter of the pixels **24** is free of the conversion units **3**, **4** in an RGGB unit so that a primary light B, preferably blue light, is emitted directly from these pixels **24**. Only one of the conversion units **3**, **4** is assigned to each of the remaining pixels **24**. First, secondary light G and second secondary light R are generated via the conversion units **3**, **4**, wherein green and red light are preferred. In the conversion units **3**, **4**, the primary light B is completely or almost completely absorbed.

**[0106]** Three pixels **24**, emitting different colors, are combined to form a display area **6**, also known as a pixel. The pixels are adapted to emit light of different colors, which is composed of the primary light B and the secondary light G, R.

**[0107]** In FIGS. 1A-1F, the individual display areas **6** each have exactly one area for red, green and blue light, thus



forming an RGB unit. There can also be two areas for green light so that an RGGb unit with four pixels **26** is formed, or an additional unit for yellow light for an RGGY unit. It is also possible to add a fluorescent unit to produce white light, resulting in an RGBW unit. White or yellow light is generated in a not shown third conversion unit stacked above the first and second conversion units **3**, **4**. The display areas **6** preferably comprise each exactly three or four of the pixels **26**. An ultraviolet-emitting semiconductor layer sequence can also be present as primary source **2** provided with three phosphors for red, blue and green and optionally for yellow or white, wherein the individual phosphor areas can be preferably excited independently of each other.

[0108] Since the conversion units **3**, **4** do not overlap at the semiconductor layer sequence **22**, the conversion units **3**, **4** can be applied in any order.

[0109] In addition to the planarization layers **73** of FIG. 1F, others not illustrated, especially transparent thin layers can also be used as additional or alternative passivation and/or encapsulation. In addition to silicon dioxide, the planarization layers **73**, the optional intermediate layer and the passivation and/or encapsulation can also be made of silicon nitride, aluminum oxide, tantalum nitride, transparent oxides or nitrides of Zn, Sn, Ta, Ga, Ni, Zr, Hf, Ti or rare earth metals. The two planarization layers **73** can be made of the same material or of different materials.

[0110] To avoid a complex adjustment between the pixels **24** of the semiconductor layer sequence **22** and the structured conversion units **3**, **4**, emission of primary radiation from the semiconductor layer sequence **22** can be used by certain actively operated pixels **24** to locally cure or solubilize photoresists. This is, for example, done when structuring the conversion units **3**, **4** directly on the semiconductor layer sequence **22** as shown in FIG. 1F.

[0111] For improved light extraction, the conversion units **3**, **4** or the semiconductor layer sequence **22** can each have a roughening on a side facing away from the carrier **21**. It may be possible to apply additional protective layers to such a roughening, which can be planarized optionally. Furthermore, the planarization layers **73** or at least one planarization layer **73** remote from the carrier **21** can be roughened optionally.

[0112] In the process of FIGS. 2A and 2B, particularly FIG. 2A, the semiconductor materials **31**, **41** are grown on two different growth substrates **5**. Subsequently, as shown in FIG. 2B, the semiconductor materials **31**, **41** are bonded together and one of the growth substrates **5** is removed.

[0113] In the process illustrated in FIGS. 3A-3C, semiconductor materials **31**, **41** are grown on separate growth substrates **5** and also structured on growth substrates **5** as shown in FIG. 3A. The planarization layers **73** are applied subsequent as shown in FIG. 3B.

[0114] Subsequently, either the two conversion units **3**, **4** are first connected and then attached to the semiconductor layer sequence **22**, or the conversion units **3**, **4** are attached sequentially to the semiconductor layer sequence **22**, as shown in FIG. 3C.

[0115] In the example of semiconductor component **1** as shown in FIG. 4, there are opaque partition walls **25a**, **25b** between adjacent pixels **24**. Optical isolation between pixels **24**, or alternatively only between adjacent display areas **6**, can be achieved via these partition walls **25a**, **25b** and

optical crosstalk is reduced or prevented. Thus, more saturated colors can be displayed and a larger gamut range can be achieved.

[0116] The partition walls **25a**, **25b** can be created by structuring the conversion units **3**, **4**, for example, by metallic or dielectric mirroring of etched flanks of the semiconductor materials **31**, **32** before generating the corresponding planarization layer **73**. The partition walls **25a**, **25b** can also be created after generating the planarization layer **73**, for example, by etching trenches and subsequent filling with reflective or absorbing material.

[0117] Partition walls **25a** can be limited to conversion units **3**, **4**. The partitions **25a** do only reach the semiconductor layer sequence **22**, but do not reach into the semiconductor layer sequence **22**. On the other hand, in partition walls **25b** generated subsequently after structuring the conversion units **3**, **4**, they can reach into the semiconductor layer sequence **22** and completely penetrate the semiconductor layer sequence **22**, optionally together with the optionally available separating regions **26**.

[0118] The partition walls **25a**, **25b**, for example, are made of a metal. If the partitions **25b** are made of an electrically conductive material, the partitions **25b** especially prefer not to reach a non-drawn metallization between the pixels **24** to avoid electrical short circuits. Alternatively, an electrical contacting of the semiconductor layer sequence **22** can be achieved via the partition walls **25a**, **25b** on a side facing away from the carrier **21**. The partition walls **25a**, **25b** can also be metal spacers or optical separations generated by photo technology.

[0119] It is possible that individual pixels are structured on 24 p sides, for example, in particular on a side facing the carrier **21**. For an n-contact, especially on a side facing away from the carrier **21**, the semiconductor layer sequence **22** between adjacent pixels **24** can be partially removed. For better optical isolation of the pixels **24** from each other, the n-conducting GaN can be removed from the n side to this n contact then, especially before the conversion units **3**, **4** are applied. All conversion units **3**, **4** can also be applied first, structured and planarized, and a trench or recesses between the individual pixels **24** is only subsequent structured, especially lithographically. These trenches or recesses filled with a reflective or absorbent material can extend into the n-GaN.

[0120] Such partition walls **25a**, **25b** can also be present in all other examples. In the semiconductor component **1**, only the partition walls **25a** of the left half of the picture in FIG. 4 or only the partition walls **25b** of the right half of the picture are preferred.

[0121] In the example of FIG. 5, additional mirror layers **71** and/or filter layers **72** are provided. The mirror layers **71**, for example, dichroic and/or dielectric mirrors are each located on a side of the associated conversion unit **3**, **4** facing the primary light source **2**. In particular, the mirror layers **71** are directly attached to the semiconductor material **31**, **41**. The mirror layers **71** ensure that the secondary light generated from the primary light does not return to the semiconductor layer sequence **22**. In contrast to FIG. 5, it is possible that the mirror layer **71** is generated continuously directly at the semiconductor layer sequence **22** and not at the first or second semiconductor material **31**, **41**.

[0122] Alternatively or in addition to the mirror layers **71**, filter layers **72** are available. The filter layers **72** prevent unconverted, in particular blue primary light from exiting

the individual pixels **24**. The filter layer **72** can be applied continuously over several of the pixels **24** or be limited to individual pixels **24**.

**[0123]** FIGS. **6A** and **6B** illustrate a method of producing the partition walls **25a**. According to FIG. **6A**, a raw material layer **27** is applied to the structured semiconductor material **31** continuously and conformally with a constant thickness. For example, the raw material layer **27** is a metal layer. The raw material layer **27** is applied, for example, by chemical vapor deposition, physical vapor deposition, atomic layer deposition or sputtering.

**[0124]** Subsequently as shown in FIG. **6B**, an anisotropic etching such as dry etching takes place so that the raw material layer **27** remains only on the flanks of the semiconductor material **31** and thus forms the partition walls **25a**.

**[0125]** The process steps of FIGS. **6A** and **6B** can, for example, be additionally performed in the process steps of FIG. **1B** or **3A** or **1F**.

**[0126]** In context to FIG. **7**, a further example is described in which pixels **24** are present that emit a first primary light **B1** without a wavelength conversion. In addition, pixels **24** are present that emit a second primary light **G2** without wavelength conversion. The first primary light **B1** is blue light, the second primary light **G2** is green light. Blue and green light are generated by electric pumping. There are also pixels **24** that emit secondary light **R1**, for example, red light. The red light is generated by the second conversion units **4**.

**[0127]** The semiconductor layer sequence **22** has a first semiconductor layer sequence **22a** that generates the second primary light **G2** and a second semiconductor layer sequence **22b** that generates the first primary light **B1**. The layer sequence directed away from the carrier **21** in the semiconductor layer sequence **22** is, for example, as follows: first semiconductor layer sequence **22a** comprising a p-doped layer, for example, p-GaN, an active layer that generates green light, an n-doped layer, for example, n-GaN, tunnel contact, second semiconductor layer sequence **22b** comprising a p-doped layer, for example, p-GaN, an active layer that generates blue light, an n-doped layer, for example, n-GaN.

**[0128]** For blue pixels **24**, the p-doped layer and the active layer that generate green light of the first semiconductor layer sequence **22a** are removed. The n contact connects to the upper n-doped layer of the second semiconductor layer sequence **22b**.

**[0129]** For green pixels **24**, the n contact connects to the lower n-doped layer of the first semiconductor layer sequence **22a**. The not shown n contact is led out laterally.

**[0130]** For red pixels **24**, the p-doped layer and the active layer that generate green light of the first semiconductor layer sequence **22a** can be removed. The n contact connects to the upper n-doped layer of the second semiconductor layer sequence **22b**. Alternatively, the p-doped layer and the active layer can remain to generate green light and can be used to excite a conversion unit **4**. In this example, the n contact connects to the lower n-doped layer of the first semiconductor layer sequence **22a**.

**[0131]** Our methods and components are not limited by the description given in the examples. Rather, this disclosure includes each new feature as well as each combination of features including in particular the combination of features in the appended claims, even if the feature or combination itself is not explicitly specified in the claims or examples.

**[0132]** Priority of DE 102016220915.9 is claimed, the subject matter of which is incorporated herein by reference.

1. A method of producing optoelectronic semiconductor components including:

providing a primary light source having a carrier and a semiconductor layer sequence mounted thereon that generates primary light (B), wherein the semiconductor layer sequence is structured into a plurality of pixels that can be driven electrically independently of each other, and the carrier comprises a plurality of control units that drive the pixels,

providing at least one conversion unit adapted to convert the primary light (B) into at least one secondary light (G, R), wherein the conversion unit is grown continuously from at least one semiconductor material,

structuring the conversion unit, wherein portions of the semiconductor material are removed in accordance with the pixels, and

applying the conversion unit to the semiconductor layer sequence so that the remaining semiconductor material is uniquely assigned to a portion of the pixels,

wherein at least two conversion units are grown on a common growth substrate, and wherein at least one of the conversion units is structured while still applied to the common growth substrate.

2. The method of producing optoelectronic semiconductor components according to claim 1, wherein the at least two conversion units are grown one above the other on a common growth substrate.

3. The method of producing optoelectronic semiconductor components according to claim 1, wherein neither structuring the semiconductor layer sequence to the pixels nor structuring the semiconductor material changes a position of remaining regions of the semiconductor layer sequence or the semiconductor material relative to each other.

4. The method of producing optoelectronic semiconductor components according to claim 1, wherein only one of the conversion units is structured on the common growth substrate, and, after structuring of this conversion unit, the semiconductor layer sequence is mounted and then detached of the growth substrate, and only subsequent the at least one further conversion unit is structured.

5. The method of producing optoelectronic semiconductor components according to claim 1, wherein at least one conversion unit is attached to the semiconductor layer sequence by wafer bonding, and a light path between the carrier and a light-exit side of the conversion unit facing away from the carrier is free of organic materials.

6. The method of producing optoelectronic semiconductor components according to claim 1, wherein some of the pixels are not assigned a conversion unit, so that the selected pixels emit the primary light (B), remaining pixels each being assigned exactly one conversion unit so that there are no conversion units stacked one above the other, and three pixels emitting different colors are combined to form a display region that is adjustable to emit light of different colors.

7. The method of producing optoelectronic semiconductor components according to claim 1, wherein the conversion units are each structured on the associated growth substrate, wherein a planarization layer made of a radiation-permeable material is applied in each case after structuring, and the planarization layer completely covers the con-

version units at least temporarily on a side facing away from the associated growth substrate.

8. The method of producing optoelectronic semiconductor components according to claim 1, wherein adjacent pixels are optically isolated by an opaque partition wall from each other, the partition wall completely penetrates at least one conversion unit and at least partially the semiconductor layer sequence.

9. The method of producing optoelectronic semiconductor components according to claim 1, wherein the semiconductor layer sequence extends continuously and contiguously over all the pixels.

10. The method of producing optoelectronic semiconductor components according to claim 1, wherein at least one mirror layer is applied to a side of the conversion units facing the semiconductor layer sequence, and the mirror layer is impermeable to the secondary light (G, R) produced in the associated conversion unit and permeable to the primary light (B).

11. The method of producing optoelectronic semiconductor components according to claim 1, wherein at least one filter layer is applied to a side of the conversion units facing away from the carrier, and the filter layer is impermeable to the primary light (B),

wherein the filter layer completely covers the conversion units.

12. The method of producing optoelectronic semiconductor components according to claim 1, wherein;

the conversion unit or each of the conversion units and/or the semiconductor layer sequence have a thickness between 1  $\mu\text{m}$  and 10  $\mu\text{m}$  inclusive,

the pixels having an average diameter between 3  $\mu\text{m}$  and 200  $\mu\text{m}$  inclusive in plan view,

wherein a distance between adjacent pixels is between 0.3  $\mu\text{m}$  and 6  $\mu\text{m}$  inclusive, and

wherein the finished semiconductor component includes between inclusive 100 and inclusive  $10^7$  of the pixels.

13. A method of producing optoelectronic semiconductor components including:

providing a primary light source having a carrier and a semiconductor layer sequence mounted thereon that generates primary light (B), wherein the semiconductor

layer sequence is structured into a plurality of pixels that can be driven electrically independently of each other, and the carrier comprises a plurality of control units that drive the pixels,

providing at least one conversion unit adapted to convert the primary light (B) into at least one secondary light (G, R), wherein the conversion unit is grown continuously from at least one semiconductor material,

structuring the conversion unit, wherein portions of the semiconductor material are removed in accordance with the pixels, and

applying the conversion unit to the semiconductor layer sequence so that the remaining semiconductor material is uniquely assigned to a portion of the pixels,

wherein the conversion unit is attached to the semiconductor layer sequence by direct bonding or anodic bonding or adhesion bonding.

14. A method of producing optoelectronic semiconductor components including:

providing a primary light source having a carrier and a semiconductor layer sequence mounted thereon that generates primary light (B), wherein the semiconductor layer sequence is structured into a plurality of pixels that can be driven electrically independently of each other, and the carrier comprises a plurality of control units that drive the pixels,

providing at least one conversion unit adapted to convert the primary light (B) into at least one secondary light (G, R), wherein the conversion unit is grown continuously from at least one semiconductor material,

structuring the conversion unit, wherein portions of the semiconductor material are removed in accordance with the pixels, and

applying the conversion unit to the semiconductor layer sequence so that the remaining semiconductor material is uniquely assigned to a portion of the pixels,

wherein adjacent pixels are optically isolated from each other.

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